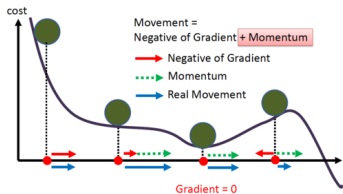
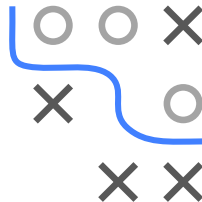


First order methods

GD with Momentum



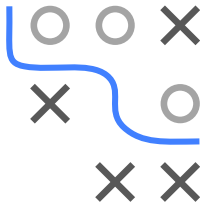
- Recap of GD problems
- Momentum definition
- Unrolling formula
- Examples
- Nesterov



RECAP: WEAKNESSES OF GRADIENT DESCENT

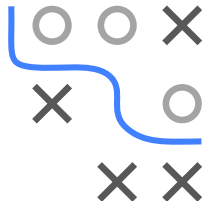
- **Zig-zagging behavior:** For ill-conditioned problems, GD moves with a zig-zag course to the optimum, since the gradient points approximately orthogonal in the shortest direction to the minimum.
- **Slow crawling:** Gradient may vanish rapidly close to stationary points (e.g. saddle points) and hence also slows down progress.
- **Trapped in stationary points:** In some functions, GD converges to stationary points (e.g. saddle points) since gradient on all sides is fairly flat and the step size is too small to pass this flat part.

Aim: More efficient algorithms which quickly reach the minimum.



GD WITH MOMENTUM: CHARACTERISTICS

- Length of a single step depends on how large and aligned a sequence of gradients is
- Length of a single step grows if many successive gradients point in the same direction
- φ determines how strongly previous gradients are included in ν
- Common values for φ are 0.5, 0.9 and even 0.99
- In general, the larger φ is in relation to α , the more strongly previous gradients influence the current direction
- **Special case** $\varphi = 0$: “vanilla” gradient descent
- **Intuition:** GD with “short term memory” for the direction of motion



MOMENTUM: ANALYSIS

$$\boldsymbol{\nu}^{[1]} = \varphi \boldsymbol{\nu}^{[0]} - \alpha \nabla f(\mathbf{x}^{[0]})$$

$$\mathbf{x}^{[1]} = \mathbf{x}^{[0]} + \varphi \mathbf{v}^{[0]} - \alpha \nabla f(\mathbf{x}^{[0]})$$

$$\boldsymbol{\nu}^{[2]} = \varphi \boldsymbol{\nu}^{[1]} - \alpha \nabla f(\mathbf{x}^{[1]})$$

$$= \varphi(\varphi \mathbf{v}^{[0]} - \alpha \nabla f(\mathbf{x}^{[0]})) - \alpha \nabla f(\mathbf{x}^{[1]})$$

$$\mathbf{x}^{[2]} = \mathbf{x}^{[1]} + \varphi(\varphi \mathbf{v}^{[0]} - \alpha \nabla f(\mathbf{x}^{[0]})) - \alpha \nabla f(\mathbf{x}^{[1]})$$

$$\mathbf{v}^{[3]} = \varphi \mathbf{v}^{[2]} - \alpha \nabla f(\mathbf{x}^{[2]})$$

$$= \varphi(\varphi(\varphi \nu^{[0]} - \alpha \nabla f(\mathbf{x}^{[0]})) - \alpha \nabla f(\mathbf{x}^{[1]})) - \alpha \nabla f(\mathbf{x}^{[2]})$$

$$\mathbf{x}^{[3]} = \mathbf{x}^{[2]} + \textcolor{teal}{\varphi}(\textcolor{teal}{\varphi}(\textcolor{teal}{\varphi}\boldsymbol{\nu}^{[0]} - \alpha\nabla f(\mathbf{x}^{[0]})) - \alpha\nabla f(\mathbf{x}^{[1]})) - \alpha\nabla f(\mathbf{x}^{[2]})$$

$$= \mathbf{x}^{[2]} + \varphi^3 \boldsymbol{\nu}^{[0]} - \varphi^2 \alpha \nabla f(\mathbf{x}^{[0]}) - \varphi \alpha \nabla f(\mathbf{x}^{[1]}) - \alpha \nabla f(\mathbf{x}^{[2]})$$

$$= \mathbf{x}^{[2]} - \alpha(\varphi^2 \nabla f(\mathbf{x}^{[0]}) + \varphi^1 \nabla f(\mathbf{x}^{[1]}) + \varphi^0 \nabla f(\mathbf{x}^{[2]})) + \varphi^3 \boldsymbol{\nu}^{[0]}$$

$$\mathbf{x}^{[t+1]} = \mathbf{x}^{[t]} - \alpha \sum_{j=0}^t \varphi^j \nabla f(\mathbf{x}^{[t-j]}) + \varphi^{t+1} \boldsymbol{\nu}^{[0]}$$

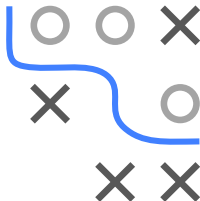
MOMENTUM: INTUITION

Suppose momentum always observes the same gradient $\nabla f(\mathbf{x}^{[t]})$:

$$\begin{aligned}\mathbf{x}^{[t+1]} &= \mathbf{x}^{[t]} - \alpha \sum_{j=0}^t \varphi^j \nabla f(\mathbf{x}^{[t-j]}) + \varphi^{t+1} \boldsymbol{\nu}^{[0]} \\ &= \mathbf{x}^{[t]} - \alpha \nabla f(\mathbf{x}^{[t]}) \sum_{j=0}^t \varphi^j + \varphi^{t+1} \boldsymbol{\nu}^{[0]} \\ &= \mathbf{x}^{[t]} - \alpha \nabla f(\mathbf{x}^{[t]}) \frac{1 - \varphi^{t+1}}{1 - \varphi} + \varphi^{t+1} \boldsymbol{\nu}^{[0]} \\ &\rightarrow \mathbf{x}^{[t]} - \alpha \nabla f(\mathbf{x}^{[t]}) \frac{1}{1 - \varphi} \quad \text{for } t \rightarrow \infty.\end{aligned}$$

Momentum accelerates along $-\nabla f(\mathbf{x}^{[t]})$ to terminal velocity yielding step size $\alpha/(1 - \varphi)$.

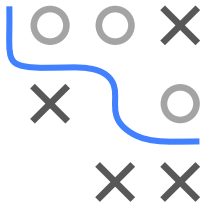
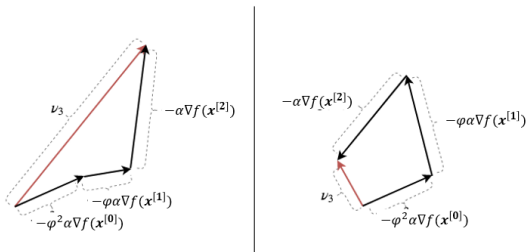
Example: Momentum with $\varphi = 0.9$ corresponds to a tenfold increase in original step size α compared to vanilla gradient descent



MOMENTUM: INTUITION (EXAMPLE)

Vector $\nu^{[3]}$ (for $\nu^{[0]} = 0$):

$$\begin{aligned}\nu^{[3]} &= \varphi(\varphi(\varphi\nu^{[0]} - \alpha\nabla f(\mathbf{x}^{[0]})) - \alpha\nabla f(\mathbf{x}^{[1]})) - \alpha\nabla f(\mathbf{x}^{[2]}) \\ &= -\varphi^2\alpha\nabla f(\mathbf{x}^{[0]}) - \varphi\alpha\nabla f(\mathbf{x}^{[1]}) - \alpha\nabla f(\mathbf{x}^{[2]})\end{aligned}$$



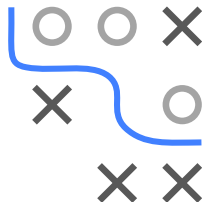
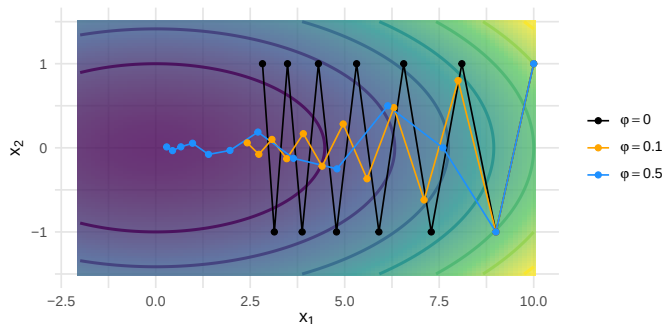
Successive gradients pointing in same/different directions increase/decrease velocity.

Further geometric intuitions and detailed explanations: [► Goh 2017](#)

GD WITH MOMENTUM: ZIG-ZAG BEHAVIOR

Consider a two-dimensional quadratic form $f(\mathbf{x}) = x_1^2/2 + 10x_2^2$.

Let $\mathbf{x}^{[0]} = (10, 1)^\top$ and $\alpha = 0.1$.

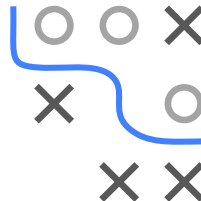
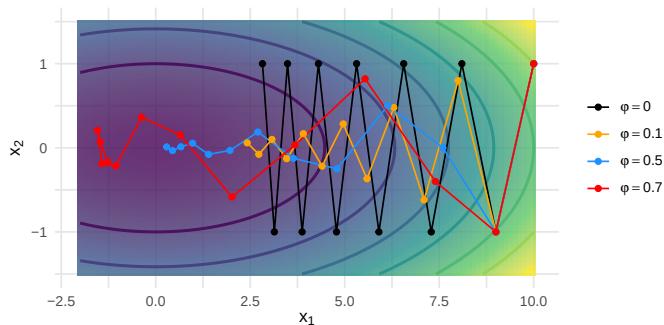


Momentum reduces zig-zagging and accelerates convergence in the direction of the minimum compared to vanilla GD.

MOMENTUM: CAUTION

Caution:

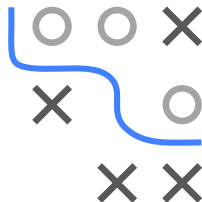
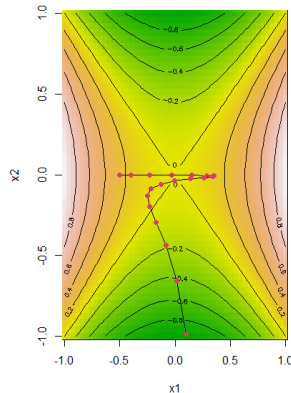
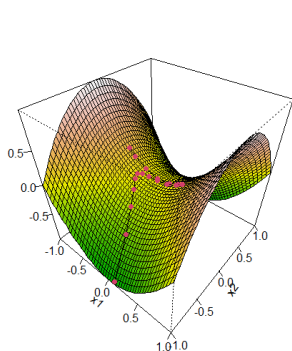
- If momentum is too high, minimum is possibly missed
- We might go back and forth around or between local minima



GD WITH MOMENTUM: SADDLE POINTS

Consider the two-dimensional quadratic form $f(\mathbf{x}) = x_1^2 - x_2^2$ with a saddle point at $(0, 0)^\top$.

Let $\mathbf{x}^{[0]} = (-1/2, 10^{-3})^\top$ and $\alpha = 0.1$.



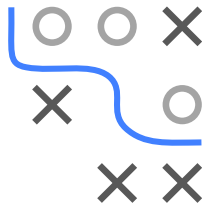
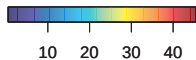
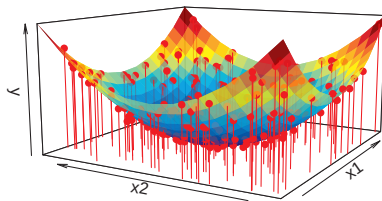
GD was slowing down at the saddle point (vanishing gradient); GD with momentum “breaks out” of the saddle point and moves on.

ERM FOR NN WITH GD

Let $\mathcal{D} = ((\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(n)}, y^{(n)}))$, with $y = x_1^2 + x_2^2$ and minimize

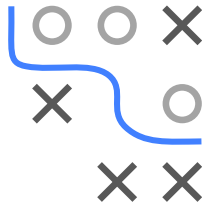
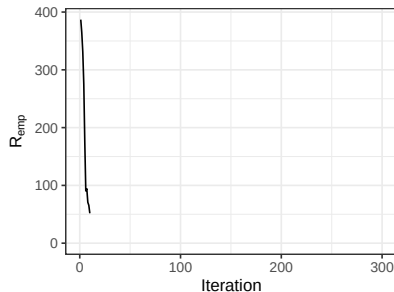
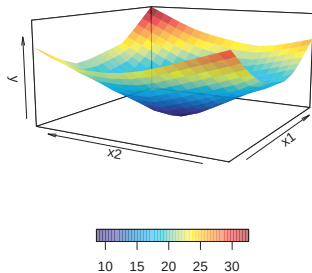
$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \sum_{i=1}^n (f(\mathbf{x} \mid \boldsymbol{\theta}) - y^{(i)})^2$$

where $f(\mathbf{x} \mid \boldsymbol{\theta})$ is a neural network with two hidden layers (16 ReLU units each).



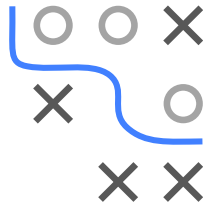
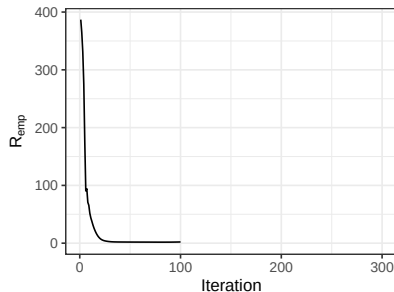
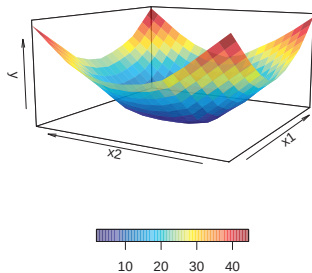
ERM FOR NN WITH GD: 10 ITERATIONS

After 10 iters of GD with momentum $\varphi = 0.5$:



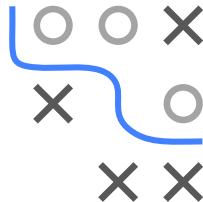
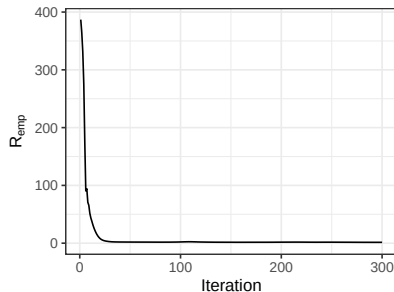
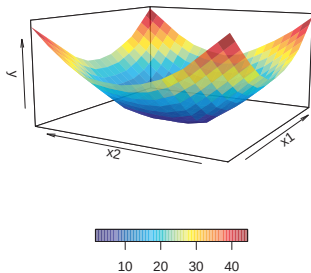
ERM FOR NN WITH GD: 100 ITERATIONS

After 100 iters of GD with momentum $\varphi = 0.5$:



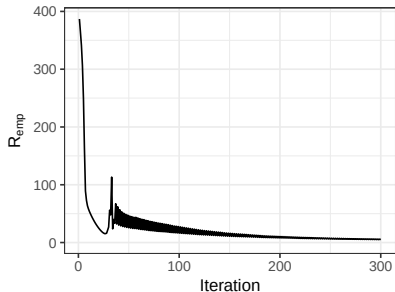
ERM FOR NN WITH GD: 300 ITERATIONS

After 300 iters of GD with momentum $\varphi = 0.5$:

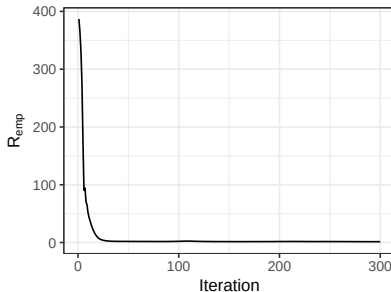


GRADIENT DESCENT WITH AND WITHOUT MOMENTUM

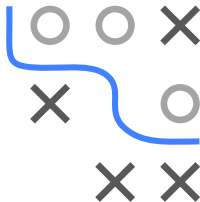
Without momentum ($\varphi = 0$):



With momentum ($\varphi = 0.5$):



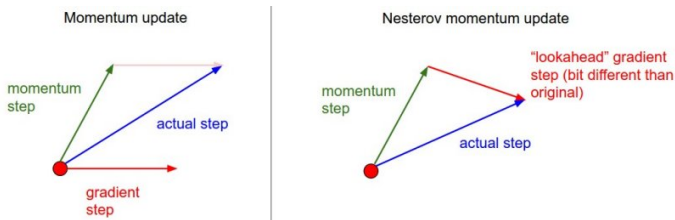
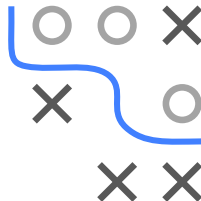
Momentum eliminates the oscillations of vanilla GD (potentially caused by overshooting in high-curvature directions with a too-large learning rate), resulting in faster, smoother convergence



NESTEROV ACCELERATED GRADIENT

- Slightly modified version: **Nesterov accelerated gradient**
- Stronger theoretical convergence guarantees for convex functions
- Avoid moving back and forth near optima

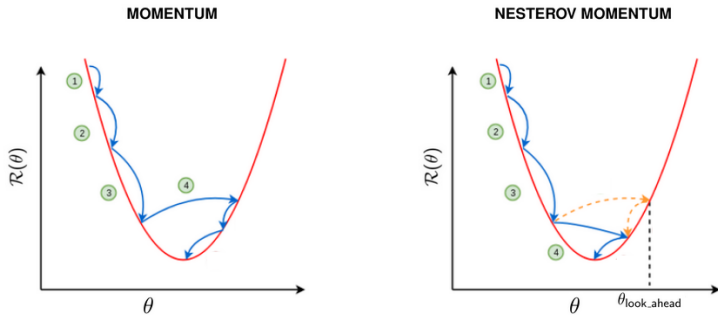
$$\begin{aligned}\nu^{[t+1]} &= \varphi \nu^{[t]} - \alpha \nabla f(\mathbf{x}^{[t]} + \varphi \nu^{[t]}) \\ \mathbf{x}^{[t+1]} &= \mathbf{x}^{[t]} + \nu^{[t+1]}\end{aligned}$$



► [Click for source](#)

Nesterov momentum update evaluates gradient at the “look-ahead” position.

MOMENTUM VS. NESTEROV



► [Click for source](#)

GD with momentum (**left**) vs. GD with Nesterov momentum (**right**).
Near minima, momentum makes a large step due to gradient history.
Nesterov mom. "looks ahead" and reduces effect of gradient history.

